

XINYI (EVELYN) GAO

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Education

University of Chicago

Bachelor of Science in Computer Science and Bachelor of Arts in Statistics (GPA: 3.8/4.0)

Expected June 2028

Chicago, IL

Technical Skills

Languages: Python, SQL, R, Java, C, TypeScript, JavaScript, HTML/CSS

Machine Learning & Data: scikit-learn, Pandas, NumPy, SciPy, GeoPandas, Matplotlib, feature engineering, PCA, clustering, time-series forecasting

AI Engineering: Model Context Protocol (MCP), FastMCP, LangChain, OpenAI Embeddings, prompt-driven tool design

Tools & Frameworks: Git, GitHub Actions (CI/CD), pytest, Docker, Pydantic, Supabase, Google Cloud Platform, React, Next.js

Coursework: Machine Learning, Data Structures & Algorithms, Linear Algebra, Biostatistics, Discrete Mathematics

Experience

Salk Institute, Harnessing Plants Initiative (Busch Lab & Talmo Lab)

Jun. 2026 – Aug. 2026

Research Software Engineering Intern

La Jolla, CA

- Shipped bloom-mcp, an MCP server of 23 validated tools (FastMCP, Pydantic) that lets AI agents run self-validating root-phenotyping analyses — data QC and cleaning, outlier detection with automatic method fallback, PCA/UMAP, clustering with quality metrics, cross-experiment statistics, and plotting — from plain-English requests in Claude.
- Designed a contract layer validating every tool's inputs and outputs and stamping each run with seeds, library versions, and SHA-256 output hashes for full reproducibility.
- Built pluggable read/write persistence across Supabase/S3 storage and a local mode for private data; replaced per-trait reads with one bulk database query.
- Reproduced published wheat analyses as golden-value regression tests in CI; delivered every tool test-first via peer-reviewed PRs.

Sony

May 2025 – Sep. 2025

Computer Science Operations & Maintenance Intern

Beijing, China

- Built Python data pipelines that prepared album and digital release assets — file retrieval, metadata cleaning, batch processing, and delivery dispatch to online music platforms — cutting release prep time ~60% and reducing errors.
- Designed a text classification system that categorized and routed 300+ user feedback comments per week, with configurable keyword and regex rules.
- Ran data quality and migration updates across a 100,000+ row production database.

Tsinghua University, Vehicle Emission Research Group

Jul. 2024 – Jan. 2025

Research Assistant, Spatiotemporal Modeling and Forecasting

Beijing, China

- Built a supervised forecasting pipeline predicting vehicle emission trajectories 24 hours ahead from 30 days of time-series data, engineering lagged and seasonal features with time-based train/validation splits to prevent leakage.
- Detected statistically significant emission hotspots using spatial autocorrelation (Moran's I, Getis-Ord Gi*) in GeoPandas, prioritizing regions for regulatory investigation.
- Communicated findings across temporal, spatial, and category-level analyses in a report adopted by the research group.

Projects

Proxima | OpenAI Embeddings, Vector Similarity, Python, Next.js, TypeScript

Jan. 2026 – Jun. 2026

- Built a semantic search engine matching students to research professors, ranking 1,500+ profile embeddings (OpenAI) by cosine similarity with LLM-generated match explanations.
- Designed the full retrieval pipeline — embedding generation, similarity scoring, reranking — and evaluated ranking quality against hand-labeled matches.
- Surfaced per-match confidence scores with relevance visualizations so users can calibrate trust in each recommendation.

AlmaBot | Python, LangChain, Gradio

May 2025

- Built a course planning system that sequences degree requirements across terms via topological sort over a prerequisite dependency graph, subject to workload constraints.
- Indexed 80,000+ course records and built LangChain LLM pipelines translating natural-language preferences into structured planning constraints, served through Gradio.